

PHYSICS

- 1) Presbyopia
- 2) Positive
- 3) Virtual, erect and same size
- 4) Below the apparent image of the fish

5) $P = 1.5 \text{ D}$

$$f = \frac{1}{P}$$

$$f = \frac{1}{1.5} \text{ m}$$

$$f = \frac{2}{3} \times 100$$

$$f = 66.7 \text{ cm}$$

$\therefore$ the power is positive, lens is conver i.e., converging

- 6) Current flowing through the circuit
 $= 5 \text{ A}$

$$\text{Time} = 30 \text{ min}$$

$$I = \frac{Q}{T}$$

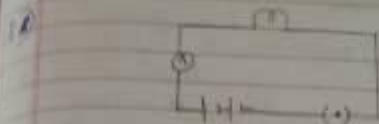
$$Q = It$$

$$Q = 5 \times 1800$$

$$Q = 9000 \text{ C}$$

$$\therefore Q = 9 \times 10^3 \text{ coulombs of charge}$$

flows through the circuit



- connecting wire
- Bulb
- Ammeter
- Battery
- Switch (closed)

2)

$$h = 5 \text{ cm}$$

$$b = 12 \text{ cm}$$

$$v = -20 \text{ cm}$$

$$\frac{1}{b} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{1}{12} = \frac{1}{v} + \frac{1}{-20}$$

$$\frac{1}{v} = \frac{1}{-20} + \frac{1}{12}$$

$$v = 30 \text{ cm}$$

$$\text{height} = h'$$

$$\frac{h'}{h} = \frac{v'}{v}$$

$$h' = \frac{30}{-20}$$

height = 2.5 cm
The image is real and inverted

9) Potential difference = $10 - 6 = 5$

$$V = \frac{W}{Q}$$

$$Q = \frac{W}{V}$$

$$Q = \frac{5}{5}$$

$$Q = 1$$

10) a) Pole :- It is the geometrical centre of the ~~sphere~~ ^{reflecting} surface of which the reflecting surface. It is represented by P.

b) Radius of curvature :- It is the radius of the sphere of which the mirror forms the part. It is represented by R.

d) Principal focus :- It is a point on the principal axis where the parallel incident rays meet after reflecting.

c) Principal axis :- A straight line passing through the pole and the centre of ~~curvature~~ curvature.

classmate
Date _____
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Q) The near point of a normal eye is 25 cm.

i) The ability of the eye lens to focus near and far objects clearly on the retina by adjusting its focal length.

ii) Since the blue colour has a shorter wave length the red colour has a longer wavelength the red colour of the eye is able to reach us after the atmospheric scattering of light.